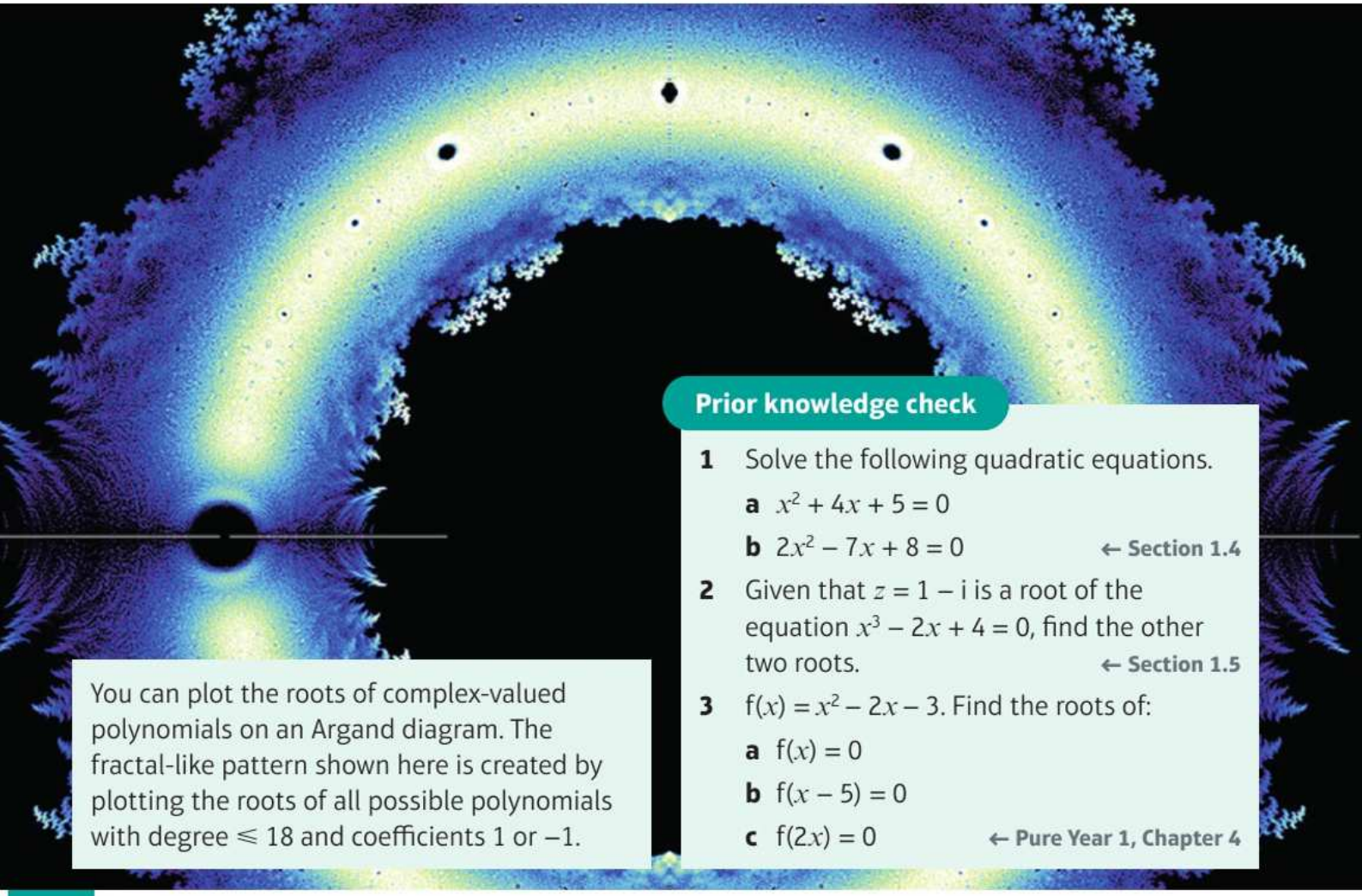
4 Roots of polynomials

Objectives

After completing this chapter you should be able to:

- Derive and use the relationships between the roots of a quadratic equation → pages 55–57
- Derive and use the relationships between the roots of a cubic equation → pages 57–59
- Derive and use the relationships between the roots of a quartic equation → pages 59–61
- Evaluate expressions relating to the roots of polynomials → pages 62–64
- Find the equation of a polynomial whose roots are a linear transformation of the roots of a given polynomial → pages 65–67



You can plot the roots of complex-valued polynomials on an Argand diagram. The fractal-like pattern shown here is created by plotting the roots of all possible polynomials with degree $\leqslant 18$ and coefficients 1 or -1.

Prior knowledge check

1 Solve the following quadratic equations.

a $x^2 + 4x + 5 = 0$

b $2x^2 - 7x + 8 = 0$ ← Section 1.4

2 Given that $z = 1 - i$ is a root of the equation $x^3 - 2x + 4 = 0$, find the other two roots. ← Section 1.5

3 $f(x) = x^2 - 2x - 3$. Find the roots of:

a $f(x) = 0$

b $f(x - 5) = 0$

c $f(2x) = 0$ ← Pure Year 1, Chapter 4

4.1 Roots of a quadratic equation

A quadratic equation of the form $ax^2 + bx + c = 0$, $x \in \mathbb{C}$, where a, b and c are real constants, can have two real roots, one repeated (real) root or two complex roots.

Links If the roots of a quadratic equation with real coefficients are complex, then they occur as a conjugate pair. ← Section 1.4

If the roots of this equation are α and β, you can determine the relationship between the **coefficients** of the terms in the quadratic equation and the values of α and β:

$$\begin{aligned} ax^2 + bx + c &= a(x - \alpha)(x - \beta) \\ &= a(x^2 - \alpha x - \beta x + \alpha\beta) \\ &= ax^2 - a(\alpha + \beta)x + a\alpha\beta \end{aligned}$$

Write the quadratic expression in factorised form, then rearrange into the form $ax^2 + bx + c$.

So $b = -a(\alpha + \beta)$ and $c = a\alpha\beta$.

- **If α and β are roots of the equation $ax^2 + bx + c = 0$, then:**
  - $\alpha + \beta = -\dfrac{b}{a}$
  - $\alpha\beta = \dfrac{c}{a}$

Note The sum of the roots is $-\frac{b}{a}$ and the product of the roots is $\frac{c}{a}$. Note that these values are real even if the roots are complex, because the sum or product of a conjugate pair is real.

Example 1

The roots of the quadratic equation $2x^2 - 5x - 4 = 0$ are α and β. Without solving the equation, find the values of:

a $\alpha + \beta$ **b** $\alpha\beta$ **c** $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$ **d** $\alpha^2 + \beta^2$

a $\alpha + \beta = \frac{5}{2}$

Use the result $\alpha + \beta = -\frac{b}{a}$

b $\alpha\beta = -2$

Use the result $\alpha\beta = \frac{c}{a}$

c $\dfrac{1}{\alpha} + \dfrac{1}{\beta} = \dfrac{\alpha + \beta}{\alpha\beta} = \dfrac{\frac{5}{2}}{-2} = -\frac{5}{4}$

d $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= \left(\frac{5}{2}\right)^2 - 2(-2) = \frac{41}{4}$

Problem-solving

Write each expression in terms of $\alpha + \beta$ and $\alpha\beta$:

$(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta \Rightarrow \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

Example 2

The roots of a quadratic equation $ax^2 + bx + c = 0$ are $\alpha = -\frac{3}{2}$ and $\beta = \frac{5}{4}$

Find integer values for a, b and c.

$\alpha + \beta = -\frac{3}{2} + \frac{5}{4} = -\frac{1}{4}$, so $-\frac{1}{4} = -\frac{b}{a}$

$\alpha\beta = -\frac{3}{2} \times \frac{5}{4} = -\frac{15}{8}$, so $-\frac{15}{8} = \frac{c}{a}$

$ax^2 + bx + c = 0$ can be written as $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$.

Use the values of $\alpha + \beta$ and $\alpha\beta$ to write down a quadratic equation with roots α and β.

$x^2 + \frac{1}{4}x - \frac{15}{8} = 0$

$8x^2 + 2x - 15 = 0$

$a = 8, b = 2, c = -15$

Any constant multiple of this equation will have roots α and β.

You could also set $a = 8$ to find integer solutions to the equations $-\frac{1}{4} = -\frac{b}{a}$ and $-\frac{15}{8} = \frac{c}{a}$

Exercise 4A

1 α and β are the roots of the quadratic equation $3x^2 + 7x - 4 = 0$. Without solving the equation, find the values of:

a $\alpha + \beta$ **b** $\alpha\beta$ **c** $\frac{1}{\alpha} + \frac{1}{\beta}$ **d** $\alpha^2 + \beta^2$

2 α and β are the roots of the quadratic equation $7x^2 - 3x + 1 = 0$. Without solving the equation, find the values of:

a $\alpha + \beta$ **b** $\alpha\beta$ **c** $\frac{1}{\alpha} + \frac{1}{\beta}$ **d** $\alpha^2 + \beta^2$

3 α and β are the roots of the quadratic equation $6x^2 - 9x + 2 = 0$. Without solving the equation, find the values of:

a $\alpha + \beta$ **b** $\alpha^2 \times \beta^2$

c $\frac{1}{\alpha} + \frac{1}{\beta}$ **d** $\alpha^3 + \beta^3$

Hint Try expanding $(\alpha + \beta)^3$.

4 The roots of a quadratic equation $ax^2 + bx + c = 0$ are $\alpha = 2$ and $\beta = -3$.
Find integer values for a, b and c.

5 The roots of a quadratic equation $ax^2 + bx + c = 0$ are $\alpha = -\frac{1}{2}$ and $\beta = -\frac{1}{3}$
Find integer values for a, b and c.

6 The roots of a quadratic equation $ax^2 + bx + c = 0$ are $\alpha = \frac{-1 + i}{2}$ and $\beta = \frac{-1 - i}{2}$
Find integer values for a, b and c.

7 One of the roots of the quadratic equation $ax^2 + bx + c = 0$ is $\alpha = -1 - 4i$.

a Write down the other root, β.

b Given that $a = 1$, find the values of b and c.

(P) **8** Given that $kx^2 + (k - 3)x - 2 = 0$, find the value of k if the sum of the roots is 4.

(P) **9** The equation $nx^2 - (16 + n)x + 256 = 0$ has real roots α and $-\alpha$. Find the value of n.

(P) **10** The roots of the equation $6x^2 + 36x + k = 0$ are reciprocals of each other. Find the value of k.

(P) **11** The equation $mx^2 + 4x + 4m = 0$ has roots of the form k and $2k$. Find the values of m and k.

(P) **12** The equation $ax^2 + 8x + c = 0$, where a and c are real constants, has roots α and α^*.

a Given that $\text{Re}(\alpha) = 2$, find the value of a.

b Given that $\text{Im}(\alpha) = 3i$, find the value of c.

P **13** The equation $4x^2 + px + q = 0$, where p and q are real constants, has roots α and α^*.

a Given that $\text{Re}(\alpha) = -3$, find the value of p.

b Given that $\text{Im}(\alpha) \neq 0$, find the range of possible values of q.

4.2 Roots of a cubic equation

A cubic equation of the form $ax^3 + bx^2 + cx + d = 0$, $x \in \mathbb{C}$, where a, b, c and d are real constants, will always have at least one real root. It will also have either two further real roots, one further repeated (real) root or two complex roots.

Links If a cubic equation with real coefficients has two complex roots, then they will occur as a conjugate pair. ← Section 1.5

If the roots of this equation are α, β and γ, you can determine the relationship between the coefficients of the terms in the cubic equation and the values of α, β and γ:

$$\begin{aligned} ax^3 + bx^2 + cx + d &= a(x - \alpha)(x - \beta)(x - \gamma) \\ &= a(x^3 - \alpha x^2 - \beta x^2 - \gamma x^2 + \alpha\beta x + \beta\gamma x + \gamma\alpha x - \alpha\beta\gamma) \\ &= ax^3 - a(\alpha + \beta + \gamma)x^2 + a(\alpha\beta + \beta\gamma + \gamma\alpha)x - a\alpha\beta\gamma \end{aligned}$$

So $b = -a(\alpha + \beta + \gamma)$, $c = a(\alpha\beta + \beta\gamma + \gamma\alpha)$ and $d = -a\alpha\beta\gamma$.

■ **If α, β and γ are roots of the equation $ax^3 + bx^2 + cx + d = 0$, then:**

- $\alpha + \beta + \gamma = -\dfrac{b}{a}$
- $\alpha\beta + \beta\gamma + \gamma\alpha = \dfrac{c}{a}$
- $\alpha\beta\gamma = -\dfrac{d}{a}$

Note As with the rule for quadratic equations, the sum of the roots is $-\frac{b}{a}$, and the sum of the products of all possible pairs of roots is $\frac{c}{a}$

Example 3

α, β and γ are the roots of the cubic equation $2x^3 + 3x^2 - 4x + 2 = 0$. Without solving the equation, find the values of:

a $\alpha + \beta + \gamma$ **b** $\alpha\beta + \beta\gamma + \gamma\alpha$ **c** $\alpha\beta\gamma$ **d** $\dfrac{1}{\alpha} + \dfrac{1}{\beta} + \dfrac{1}{\gamma}$

a $\alpha + \beta + \gamma = -\frac{3}{2}$ — Use the result $\alpha + \beta + \gamma = -\dfrac{b}{a}$

b $\alpha\beta + \beta\gamma + \gamma\alpha = -2$ — Use the result $\alpha\beta + \beta\gamma + \gamma\alpha = \dfrac{c}{a}$

c $\alpha\beta\gamma = -1$ — Use the result $\alpha\beta\gamma = -\dfrac{d}{a}$

d $\dfrac{1}{\alpha} + \dfrac{1}{\beta} + \dfrac{1}{\gamma} = \dfrac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \dfrac{\frac{c}{a}}{-\frac{d}{a}}$

$= \dfrac{-2}{-1} = 2$

Notice that the numerator is just the sum of the products of pairs of roots with the terms in a different order.

Example 4

The roots of a cubic equation $ax^3 + bx^2 + cx + d = 0$ are $\alpha = 1 - 2i$, $\beta = 1 + 2i$ and $\gamma = 2$.
Find integer values for a, b, c and d.

$\alpha + \beta + \gamma = (1 - 2i) + (1 + 2i) + 2 = 4$,
so $4 = -\frac{b}{a}$
$\alpha\beta + \beta\gamma + \gamma\alpha$
$= (1 - 2i)(1 + 2i) + (1 + 2i) \times 2 + 2(1 - 2i)$
$= 9$
so $9 = \frac{c}{a}$
$\alpha\beta\gamma = (1 - 2i)(1 + 2i) \times 2 = 10$,
so $10 = -\frac{d}{a}$
$x^3 - 4x^2 + 9x - 10 = 0$
$a = 1, b = -4, c = 9, d = -10$

$ax^3 + bx^2 + cx + d = 0$ can be written as $x^3 + \frac{b}{a}x^2 + \frac{c}{a}x + \frac{d}{a} = 0$.
Use the values of $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$ to write down a cubic equation with roots α, β and γ.

Watch out Be careful with negative signs when writing out the equation.

You could also set $a = 1$ to find integer solutions to the equations $4 = -\frac{b}{a}$, $9 = \frac{c}{a}$ and $10 = -\frac{d}{a}$

Exercise 4B

1 α, β and γ are the roots of the cubic equation $2x^3 + 5x^2 - 2x + 3 = 0$. Find the values of:

a $\alpha + \beta + \gamma$ **b** $\alpha\beta\gamma$ **c** $\alpha\beta + \beta\gamma + \gamma\alpha$ **d** $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

2 α, β and γ are the roots of the cubic equation $x^3 + 5x^2 + 17x + 13 = 0$. Find the values of:

a $\alpha + \beta + \gamma$ **b** $\alpha\beta\gamma$ **c** $\alpha\beta + \beta\gamma + \gamma\alpha$ **d** $\alpha^2\beta^2\gamma^2$

3 α, β and γ are the roots of the cubic equation $7x^3 - 4x^2 - x + 6 = 0$. Find the values of:

a $\alpha + \beta + \gamma$ **b** $\alpha\beta\gamma$ **c** $\alpha^3\beta^3\gamma^3$ **d** $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

4 The roots of a cubic equation $ax^3 + bx^2 + cx + d = 0$ are $\alpha = \frac{3}{2}$, $\beta = \frac{1}{2}$ and $\gamma = 1$.
Find integer values for a, b, c and d.

5 The roots of a cubic equation $ax^3 + bx^2 + cx + d = 0$ are $\alpha = 1 + 3i$, $\beta = 1 - 3i$ and $\gamma = \frac{1}{2}$
Find integer values for a, b, c and d.

6 The roots of a cubic equation $ax^3 + bx^2 + cx + d = 0$ are $\alpha = \frac{5}{4}$, $\beta = -\frac{3}{2}$ and $\gamma = \frac{1}{2}$
Find integer values for a, b, c and d.

E **7** The cubic equation $16x^3 - kx^2 + 1 = 0$ has roots α, β and γ.

a Write down the values of $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$. **(2 marks)**

b i Given that $\alpha = \beta$, find the roots of the equation. **(5 marks)**

ii Find the value of k. **(1 mark)**

E **8** The cubic equation $2x^3 - kx^2 + 30x - 13 = 0$ has roots α, β and γ.

a Write down the values of $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$, and express k in terms of α, β and γ. **(3 marks)**

b Given that $\alpha = 2 - 3i$, find the value of k. **(4 marks)**

E **9** The cubic equation $x^3 - mx + n = 0$ has roots 1, −4 and α.

a State, with a reason, whether α is real. **(1 mark)**

b Find the values of m, n and α. **(4 marks)**

E **10** The cubic equation $2x^3 - 10x^2 + 8x - k = 0$ has a root at $x = 3 - \mathrm{i}$.

a Find the other two roots of the equation. **(4 marks)**

b Hence find the value of k. **(2 marks)**

E/P **11** The cubic equation $x^3 - 14x^2 + 56x - 64 = 0$ has roots α, $k\alpha$ and $k^2\alpha$ for some real constant k. Find the values of α and k. **(5 marks)**

E/P **12** Given that the roots of $8x^3 + 12x^2 - cx + d = 0$ are α, $\frac{\alpha}{2}$ and $\alpha - 4$, find α, c and d. **(5 marks)**

E/P **13** Given that the roots of the cubic equation $2x^3 + 48x^2 + cx + d = 0$ are α, 2α and 3α, find the values of α, c and d. **(5 marks)**

Challenge

α, β and γ are the roots of the equation $ax^3 + bx^2 + cx + d = 0$, $x \in \mathbb{C}$, $a, b, c, d \in \mathbb{R}$.

By considering all the possible cases in which α, β and γ are real or complex, explain why the following are always real numbers:

a $\alpha + \beta + \gamma$ **b** $\alpha\beta + \beta\gamma + \gamma\alpha$ **c** $\alpha\beta\gamma$

4.3 Roots of a quartic equation

Consider the quartic equation $ax^4 + bx^3 + cx^2 + dx + e = 0$, $x \in \mathbb{C}$, where a, b, c, d and e are real numbers. If the roots of the equation are α, β, γ and δ, you can determine the relationship between the coefficients of the terms in the equation and the values of α, β, γ and δ:

$$\begin{aligned} ax^4 + bx^3 + cx^2 + dx + e &= a(x-\alpha)(x-\beta)(x-\gamma)(x-\delta) \\ &= a(x^4 - \alpha x^3 - \beta x^3 - \gamma x^3 - \delta x^3 + \alpha\beta x^2 + \beta\gamma x^2 + \gamma\alpha x^2 + \gamma\delta x^2 + \alpha\delta x^2 \\ &\quad + \beta\delta x^2 - \alpha\beta\gamma x - \alpha\beta\delta x - \alpha\gamma\delta x - \beta\gamma\delta x + \alpha\beta\gamma\delta) \\ &= ax^4 - a(\alpha+\beta+\gamma+\delta)x^3 + a(\alpha\beta + \beta\gamma + \gamma\alpha + \gamma\delta + \alpha\delta + \beta\delta)x^2 \\ &\quad - a(\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta)x + a\alpha\beta\gamma\delta \end{aligned}$$

So $b = -a(\alpha + \beta + \gamma + \delta)$, $c = a(\alpha\beta + \beta\gamma + \gamma\alpha + \gamma\delta + \alpha\delta + \beta\delta)$, $d = -a(\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta)$ and $e = a\alpha\beta\gamma\delta$.

■ **If α, β, γ and δ are roots of the equation $ax^4 + bx^3 + cx^2 + dx + e = 0$, then:**

- $\alpha + \beta + \gamma + \delta = -\frac{b}{a}$
- $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{c}{a}$
- $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -\frac{d}{a}$
- $\alpha\beta\gamma\delta = \frac{e}{a}$

Notation You can use the following abbreviations for these results in your working:

$\Sigma\alpha = -\frac{b}{a}$ $\Sigma\alpha\beta = \frac{c}{a}$ $\Sigma\alpha\beta\gamma = -\frac{d}{a}$

Example 5

The equation $x^4 + 2x^3 + px^2 + qx - 60 = 0$, $x \in \mathbb{C}$, $p, q \in \mathbb{R}$, has roots α, β, γ and δ.
Given that $\gamma = -2 + 4i$ and $\delta = \gamma^*$,

a show that $\alpha + \beta - 2 = 0$ and that $\alpha\beta + 3 = 0$.

b Hence find all the roots of the quartic equation and find the values of p and q.

a $\alpha + \beta + \gamma + \delta = -2$

$\alpha + \beta + (-2 + 4i) + (-2 - 4i) = -2$

$\alpha + \beta - 4 = -2$

Hence $\alpha + \beta - 2 = 0$ (1)

Use $\Sigma\alpha = -\frac{b}{a}$ with $b = 2$ and $a = 1$.

$\alpha\beta\gamma\delta = -60$

$\alpha\beta(-2 + 4i)(-2 - 4i) = -60$

$20\alpha\beta = -60$

Hence $\alpha\beta + 3 = 0$ (2)

Use $\alpha\beta\gamma\delta = \frac{e}{a}$ with $e = -60$ and $a = 1$.

You can find $(-2 + 4i)(-2 - 4i)$ quickly by remembering that $(a + bi)(a - bi) = a^2 + b^2$.

b Solve equations (1) and (2) simultaneously.

From (1), $\beta = 2 - \alpha$ so substitute into (2):

$\alpha(2 - \alpha) + 3 = 0$

$\alpha^2 - 2\alpha - 3 = 0$

$(\alpha - 3)(\alpha + 1) = 0$

$\alpha = 3$ or -1

If $\alpha = 3$, $\beta = -1$ and if $\alpha = -1$, $\beta = 3$

So the roots of the quartic equation are 3, -1, $-2 + 4i$ and $-2 - 4i$.

$\Sigma\alpha\beta = \frac{p}{a}$

Use $\Sigma\alpha\beta = \frac{c}{a}$ with $c = p$ and $a = 1$.

$\Rightarrow p = 3(-1) + 3(-2 + 4i) + 3(-2 - 4i) - (-2 + 4i) - (-2 - 4i) + (-2 + 4i)(-2 - 4i)$

So $p = 9$

$\Sigma\alpha\beta\gamma = -\frac{q}{a}$

$\Sigma\alpha\beta\gamma = -\frac{d}{a}$ with $d = q$ and $a = 1$.

$\Rightarrow -q = 3(-1)(-2 + 4i) + 3(-1)(-2 - 4i) + 3(-2 + 4i)(-2 - 4i) - (-2 + 4i)(-2 - 4i)$

So $q = -52$

Exercise 4C

1 α, β, γ and δ are the roots of the quartic equation $4x^4 + 3x^3 + 2x^2 - 5x - 4 = 0$. Without solving the equation, find the values of:

a $\alpha + \beta + \gamma + \delta$

b $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$

c $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$

d $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$

Hint $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = \frac{\beta\gamma\delta + \alpha\gamma\delta + \alpha\beta\delta + \alpha\beta\gamma}{\alpha\beta\gamma\delta}$

2 α, β, γ and δ are the roots of the quartic equation $2x^4 + 4x^3 - 3x^2 - x + 2 = 0$. Find the values of:

a $\alpha + \beta + \gamma + \delta$

b $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$

c $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$

d $\alpha\beta\gamma\delta$

e $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$

3 α, β, γ and δ are the roots of the quartic equation $x^4 + 3x^3 + 2x^2 - x + 4 = 0$.
Find the values of:

a $\alpha + \beta + \gamma + \delta$ **b** $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$ **c** $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$

d $\alpha\beta\gamma\delta$ **e** $\alpha^2\beta^2\gamma^2\delta^2$

4 α, β, γ and δ are the roots of the quartic equation $7x^4 + 6x^3 - 5x^2 + 4x + 3 = 0$.
Find the values of:

a $\alpha + \beta + \gamma + \delta$ **b** $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$ **c** $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$

d $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$ **e** $\alpha^3\beta^3\gamma^3\delta^3$

5 The roots of a quartic equation $ax^4 + bx^3 + cx^2 + dx + e = 0$ are $\alpha = -\frac{3}{2}$, $\beta = -\frac{1}{2}$, $\gamma = -2$ and $\delta = \frac{2}{3}$
Find integer values for a, b, c, d and e.

6 The roots of a quartic equation $ax^4 + bx^3 + cx^2 + dx + e = 0$ are $\alpha = -\frac{1}{2}$, $\beta = \frac{1}{3}$, $\gamma = 1 + \mathrm{i}$ and $\delta = 1 - \mathrm{i}$. Find integer values for a, b, c, d and e.

7 The roots of a quartic equation $ax^4 + bx^3 + cx^2 + dx + e = 0$ are such that $\Sigma\alpha = \frac{17}{12}$, $\Sigma\alpha\beta = -\frac{25}{72}$, $\Sigma\alpha\beta\gamma = -\frac{53}{72}$ and $\alpha\beta\gamma\delta = -\frac{1}{6}$. Find integer values for a, b, c, d and e.

E/P **8** The quartic equation $x^4 - 16x^3 + 86x^2 - 176x + 105 = 0$ has roots α, $\alpha + k$, $\alpha + 2k$ and $\alpha + 3k$ for some real constant k. Solve the equation. **(7 marks)**

E/P **9** The quartic equation $3072x^4 - 2880x^3 + 840x^2 - 90x + 3 = 0$ has roots α, $r\alpha$, $r^2\alpha$ and $r^3\alpha$ for some real constant r. Solve the equation. **(7 marks)**

E **10** Three of the roots of the quartic equation $40x^4 + 90x^3 - 115x^2 + mx + n = 0$ are 1, -3 and $\frac{1}{2}$

a Find the fourth root. **(2 marks)**

b Find the values of m and n. **(4 marks)**

E/P **11** The quartic equation $2x^4 - 34x^3 + 202x^2 + dx + e = 0$ has roots α, $\alpha + 1$, $2\alpha + 1$ and $3\alpha + 1$.

a Find α. **(2 marks)**

b Find the values of d and e. **(4 marks)**

E/P **12** The equation $4x^4 - 19x^3 + px^2 + qx + 10 = 0$, $x \in \mathbb{C}$, $p, q \in \mathbb{R}$, has roots α, β, γ and δ.
Given that $\gamma = 3 + \mathrm{i}$ and $\delta = \gamma^*$,

a show that $4\alpha + 4\beta + 5 = 0$ and that $4\alpha\beta - 1 = 0$. **(2 marks)**

b Hence find all the roots of the quartic equation and find the values of p and q. **(5 marks)**

c Show these roots on an Argand diagram. **(3 marks)**

E/P **13** A quartic equation $6x^4 - 10x^3 + 3x^2 + 6x - 40 = 0$ has roots α, β, γ and δ.

a Show that $\frac{1 - 3\mathrm{i}}{2}$ is one root of the equation. **(3 marks)**

b Without solving the equation, find the other roots. **(5 marks)**

c Show these roots on an Argand diagram. **(3 marks)**

4.4 Expressions relating to the roots of a polynomial

You have already seen several results for finding the values of expressions relating to the roots of a polynomial.

- **The rules for reciprocals:**
  - **Quadratic:** $\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}$
  - **Cubic:** $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{\alpha\beta+\beta\gamma+\gamma\alpha}{\alpha\beta\gamma}$
  - **Quartic:** $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta}=\frac{\alpha\beta\gamma+\beta\gamma\delta+\gamma\delta\alpha+\delta\alpha\beta}{\alpha\beta\gamma\delta}$
- **The rules for products of powers:**
  - **Quadratic:** $\alpha^n\times\beta^n=(\alpha\beta)^n$
  - **Cubic:** $\alpha^n\times\beta^n\times\gamma^n=(\alpha\beta\gamma)^n$
  - **Quartic:** $\alpha^n\times\beta^n\times\gamma^n\times\delta^n=(\alpha\beta\gamma\delta)^n$

In addition to these you have also used the following results for the roots of quadratic equations:

- $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$
- $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$

There are equivalent results to these for the roots of cubic and quartic equations.

Example 6

a Expand $(\alpha+\beta+\gamma)^2$.

b A cubic equation has roots α, β, γ such that $\alpha\beta+\beta\gamma+\gamma\alpha=7$ and $\alpha+\beta+\gamma=-3$. Find the value of $\alpha^2+\beta^2+\gamma^2$.

a $(\alpha+\beta+\gamma)^2=(\alpha+\beta+\gamma)(\alpha+\beta+\gamma)$
$=\alpha^2+\alpha\beta+\alpha\gamma+\beta\alpha+\beta^2+\beta\gamma+\gamma\alpha+\gamma\beta+\gamma^2$
$=\alpha^2+\beta^2+\gamma^2+2(\alpha\beta+\beta\gamma+\gamma\alpha)$

b $\alpha^2+\beta^2+\gamma^2=(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma\alpha)$
$=(-3)^2-2(7)=-5$

Rearrange the result from part **a**. You know the value of $\alpha+\beta+\gamma$ and the value of $\alpha\beta+\beta\gamma+\gamma\alpha$, so you have enough information to find $\alpha^2+\beta^2+\gamma^2$.

You can find an expression for the sum of the squares of a quartic equation in a similar way, by multiplying out $(\alpha+\beta+\gamma+\delta)^2$.

- **The rules for sums of squares:**
  - **Quadratic:** $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$
  - **Cubic:** $\alpha^2+\beta^2+\gamma^2=(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma\alpha)$
  - **Quartic:** $\alpha^2+\beta^2+\gamma^2+\delta^2=(\alpha+\beta+\gamma+\delta)^2-2(\alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\beta\delta+\gamma\delta)$

Note If you learn these you can use them without proof in your exam.

You can find a similar result for the sum of the cubes of a cubic equation by multiplying out $(\alpha + \beta + \gamma)^3$.

■ **The rules for sums of cubes:**

- **Quadratic:** $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
- **Cubic:** $\alpha^3 + \beta^3 + \gamma^3 = (\alpha + \beta + \gamma)^3 - 3(\alpha + \beta + \gamma)(\alpha\beta + \beta\gamma + \gamma\alpha) + 3\alpha\beta\gamma$

Note The result for the sum of cubes for a quartic equation is not required.

Example 7

The three roots of a cubic equation are α, β and γ. Given that $\alpha\beta\gamma = 4$, $\alpha\beta + \beta\gamma + \gamma\alpha = -5$ and $\alpha + \beta + \gamma = 3$, find the value of $(\alpha + 3)(\beta + 3)(\gamma + 3)$.

$(\alpha + 3)(\beta + 3)(\gamma + 3)$
$= \alpha\beta\gamma + 3\alpha\beta + 3\alpha\gamma + 9\alpha + 3\beta\gamma + 9\beta + 9\gamma + 27$
$= \alpha\beta\gamma + 3(\alpha\beta + \beta\gamma + \alpha\gamma) + 9(\alpha + \beta + \gamma) + 27$
$= 4 + 3(-5) + 9(3) + 27$
$= 43$

Expand the brackets.

Group the terms and factorise to write the expression in terms of the expressions given in the question.

Exercise 4D

1 A quadratic equation has roots α and β. Given that $\alpha + \beta = 4$ and $\alpha\beta = 3$, find:

a $\frac{1}{\alpha} + \frac{1}{\beta}$ **b** $\alpha^2\beta^2$ **c** $\alpha^2 + \beta^2$ **d** $\alpha^3 + \beta^3$

2 A quadratic equation has roots α and β. Given that $\alpha + \beta = -\frac{2}{3}$ and $\alpha\beta = \frac{3}{4}$, find:

a $\frac{1}{\alpha} + \frac{1}{\beta}$ **b** $\alpha^2\beta^2$ **c** $\alpha^2 + \beta^2$ **d** $\alpha^3 + \beta^3$

(P) **3** A quadratic equation has roots α and β. Given that $\alpha + \beta = \frac{5}{4}$ and $\alpha\beta = -\frac{1}{3}$, find:

a $(\alpha + 2)(\beta + 2)$ **b** $(\alpha - 4)(\beta - 4)$ **c** $(\alpha^2 + 1)(\beta^2 + 1)$

4 A cubic equation has roots α, β and γ. Given that $\alpha + \beta + \gamma = 2$, $\alpha\beta + \beta\gamma + \gamma\alpha = -3$ and $\alpha\beta\gamma = 4$, find:

a $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ **b** $\alpha^2 + \beta^2 + \gamma^2$ **c** $\alpha^3 + \beta^3 + \gamma^3$ **d** $(\alpha\beta)^2 + (\beta\gamma)^2 + (\gamma\alpha)^2$

5 A cubic equation has roots α, β and γ. Given that $\Sigma\alpha = \frac{3}{2}$, $\Sigma\alpha\beta = -\frac{4}{3}$ and $\alpha\beta\gamma = \frac{1}{2}$, find:

a $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ **b** $\alpha^2 + \beta^2 + \gamma^2$ **c** $\alpha^3 + \beta^3 + \gamma^3$ **d** $\alpha^3\beta^3\gamma^3$

(P) **6** A cubic equation has roots α, β and γ. Given that $\alpha + \beta + \gamma = -\frac{1}{2}$, $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{3}{4}$ and $\alpha\beta\gamma = -\frac{2}{5}$, find:

a $(\alpha + 2)(\beta + 2)(\gamma + 2)$ **b** $(\alpha - 3)(\beta - 3)(\gamma - 3)$ **c** $(1 - \alpha)(1 - \beta)(1 - \gamma)$

d $(\alpha\beta)^2 + (\beta\gamma)^2 + (\gamma\alpha)^2$ **e** $(\alpha\beta)^3 + (\beta\gamma)^3 + (\gamma\alpha)^3$

7 A quartic equation has roots α, β, γ and δ. Given that $\alpha + \beta + \gamma + \delta = 3$, $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = 5$, $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -4$ and $\alpha\beta\gamma\delta = -2$, find:

a $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$ **b** $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ **c** $\alpha^4\beta^4\gamma^4\delta^4$

8 A quartic equation has roots α, β, γ and δ. Given that $\Sigma\alpha = \frac{1}{2}$, $\Sigma\alpha\beta = -\frac{3}{4}$, $\Sigma\alpha\beta\gamma = -\frac{1}{5}$ and $\alpha\beta\gamma\delta = \frac{4}{3}$, find:

a $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$ **b** $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ **c** $\alpha^3\beta^3\gamma^3\delta^3$

d $(\alpha\beta)^2 + (\beta\gamma)^2 + (\gamma\alpha)^2 + (\gamma\delta)^2 + (\alpha\delta)^2 + (\beta\delta)^2$

e $(\alpha\beta\gamma)^2 + (\alpha\beta\delta)^2 + (\alpha\gamma\delta)^2 + (\beta\gamma\delta)^2$

(P) **9** A quartic equation has roots α, β, γ and δ. Given that $\Sigma\alpha = -\frac{1}{2}$, $\Sigma\alpha\beta = -\frac{1}{3}$, $\Sigma\alpha\beta\gamma = \frac{1}{4}$ and $\alpha\beta\gamma\delta = \frac{3}{2}$, find:

a $(\alpha + 1)(\beta + 1)(\gamma + 1)(\delta + 1)$ **b** $(2 - \alpha)(2 - \beta)(2 - \gamma)(2 - \delta)$

(E/P) **10** The roots of the equation $x^3 - 6x^2 + 9x - 15 = 0$ are α, β and γ.

a Write down the values of $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$. **(1 mark)**

b Hence find the values of:

i $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ **(2 marks)**

ii $\alpha^2 + \beta^2 + \gamma^2$ **(2 marks)**

iii $(\alpha - 1)(\beta - 1)(\gamma - 1)$ **(3 marks)**

(E/P) **11** The roots of the equation $2x^3 + 4x^2 + 7 = 0$ are α, β and γ.

a Write down the values of $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$. **(1 mark)**

b Hence find the values of:

i $\alpha^2 + \beta^2 + \gamma^2$ **(2 marks)**

ii $\alpha^3\beta^3\gamma^3$ **(2 marks)**

iii $(\alpha + 2)(\beta + 2)(\gamma + 2)$ **(3 marks)**

(P) **12** Show that $\alpha^3 + \beta^3 + \gamma^3 \equiv (\alpha + \beta + \gamma)^3 - 3(\alpha + \beta + \gamma)(\alpha\beta + \beta\gamma + \gamma\alpha) + 3\alpha\beta\gamma$.

(E/P) **13** The roots of the equation $3x^3 - px + 11 = 0$ are α, β and γ.

a Given that $\alpha\beta + \beta\gamma + \gamma\alpha = 4$, write down the value of p. **(1 mark)**

b Write down the values of $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$. **(1 mark)**

c Hence find the value of $(3 - \alpha)(3 - \beta)(3 - \gamma)$. **(3 marks)**

(E/P) **14** The roots of the equation $x^4 + 2x^2 - x + 3 = 0$ are α, β, γ and δ.

a Write down the values of $\Sigma\alpha$, $\Sigma\alpha\beta$, $\Sigma\alpha\beta\gamma$ and $\alpha\beta\gamma\delta$. **(1 mark)**

b Hence find the values of:

i $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$ **(3 marks)**

ii $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ **(3 marks)**

iii $(\alpha + 1)(\beta + 1)(\gamma + 1)(\delta + 1)$ **(3 marks)**

(E) **15** The roots of the equation $ax^4 + 3x^3 + 2x^2 + x - 6 = 0$ are α, β, γ and δ.

a Given that $\alpha\beta\gamma\delta = -3$, write down the value of a. **(1 mark)**

b Write down the values of $\Sigma\alpha$, $\Sigma\alpha\beta$ and $\Sigma\alpha\beta\gamma$. **(1 mark)**

c Hence find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$ **(3 marks)**

(P) **16** Prove that if a quartic equation has roots α, β, γ and δ then $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 \equiv (\Sigma\alpha)^2 - 2\Sigma\alpha\beta$.

4.5 Linear transformations of roots

Given the sums and products of the roots of a polynomial, it is possible to find the equation of a second polynomial whose roots are a linear transformation of the roots of the first.

For example, if the roots of a cubic equation are α, β and γ, you need to be able to find the equation of a polynomial with roots $(\alpha + 2)$, $(\beta + 2)$ and $(\gamma + 2)$, or 3α, 3β and 3γ.

Example 8

The cubic equation $x^3 - 2x^2 + 3x - 4 = 0$ has roots α, β and γ. Find the equations of the polynomials with roots:

a 2α, 2β and 2γ **b** $(\alpha + 3)$, $(\beta + 3)$ and $(\gamma + 3)$

a **Method 1**

$\alpha + \beta + \gamma = 2$, $\alpha\beta + \beta\gamma + \gamma\alpha = 3$ and $\alpha\beta\gamma = 4$

Sum $= 2\alpha + 2\beta + 2\gamma = 2(\alpha + \beta + \gamma) = 4$

Pair sum $= (2\alpha)(2\beta) + (2\beta)(2\gamma) + (2\gamma)(2\alpha)$
$= 4(\alpha\beta + \beta\gamma + \gamma\alpha) = 12$

Product $= (2\alpha)(2\beta)(2\gamma) = 8\alpha\beta\gamma = 32$

Hence the new equation is $w^3 - 4w^2 + 12w - 32 = 0$

Set $a = 1$ in the new equation.

Method 2

Let $w = 2x$, hence $x = \frac{w}{2}$

This result could have been derived by direct substitution. Each root in the new equation is twice the corresponding root in the original equation, so set $w = 2x$.

Substituting: $\left(\frac{w}{2}\right)^3 - 2\left(\frac{w}{2}\right)^2 + 3\left(\frac{w}{2}\right) - 4 = 0$

Multiply through by 8: $w^3 - 4w^2 + 12w - 32 = 0$

You could leave this in the above form, or multiply through by 8 to get an equation with integer coefficients.

b **Method 1**

Sum $= (\alpha + 3) + (\beta + 3) + (\gamma + 3) = \alpha + \beta + \gamma + 9 = 11$

Pair sum $= (\alpha + 3)(\beta + 3) + (\beta + 3)(\gamma + 3) + (\gamma + 3)(\alpha + 3)$
$= \alpha\beta + \beta\gamma + \gamma\alpha + 6(\alpha + \beta + \gamma) + 27$
$= 3 + 12 + 27$
$= 42$

Product $= (\alpha + 3)(\beta + 3)(\gamma + 3)$
$= \alpha\beta\gamma + 3(\alpha\beta + \beta\gamma + \gamma\alpha) + 9(\alpha + \beta + \gamma) + 27$
$= 4 + 9 + 18 + 27 = 58$

Follow the same steps. Use the results $\Sigma\alpha = 2$, $\Sigma\alpha\beta = 3$ and $\alpha\beta\gamma = 4$ from part **a**.

Hence the new equation is $w^3 - 11w^2 + 42w - 58 = 0$

Method 2

Let $w = x + 3$ hence $x = w - 3$.

Each root in the new equation is 3 more than the equivalent root in the old equation.

Substituting: $(w - 3)^3 - 2(w - 3)^2 + 3(w - 3) - 4 = 0$

$w^3 - 9w^2 + 27w - 27 - 2(w^2 - 6w + 9) + 3w - 9 - 4 = 0$

$w^3 - 11w^2 + 42w - 58 = 0$

Multiply out and simplify.

Problem-solving

Find the sum $\Sigma\alpha$, the pair sum $\Sigma\alpha\beta$ and the product $\alpha\beta\gamma$ for the original equation. Then use these values to find the equivalent sums and products for an equation with roots 2α, 2β and 2γ.

Watch out It's a good idea to choose a different letter such as w for the variable in your new equation.

Example 9

The quartic equation $x^4 - 3x^3 + 15x + 1 = 0$ has roots α, β, γ and δ. Find the equation with roots $(2\alpha + 1)$, $(2\beta + 1)$, $(2\gamma + 1)$ and $(2\delta + 1)$.

Method 1

$\alpha + \beta + \gamma + \delta = 3$

$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = 0$

$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -15$

$\alpha\beta\gamma\delta = 1$

Sum of roots:

$(2\alpha + 1) + (2\beta + 1) + (2\gamma + 1) + (2\delta + 1)$

$= 2(a + \beta + \gamma + \delta) + 4$

$= 10$

Pair sum:

$4(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) + 6(\alpha + \beta + \gamma + \delta) + 6$

$= 4 \times 0 + 6 \times 3 + 6 = 24$

Triple sum:

$8(\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta) + 8(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$
$+ 6(\alpha + \beta + \gamma + \delta) + 4 = -98$

Product:

$16\alpha\beta\gamma\delta + 8(\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta) + 4(\alpha\beta + \alpha\gamma$
$+ \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) + 2(\alpha + \beta + \gamma + \delta) + 1$

$= -97$

Hence the new equation is

$w^4 - 10w^3 + 24w^2 + 98w - 97 = 0$

Method 2

Let $w = 2x + 1$ hence $x = \frac{w - 1}{2}$

Substituting: $\left(\frac{w - 1}{2}\right)^4 - 3\left(\frac{w - 1}{2}\right)^3 + 15\left(\frac{w - 1}{2}\right) + 1 = 0$

$(w - 1)^4 - 6(w - 1)^3 + 120(w - 1) + 16 = 0$

$w^4 - 4w^3 + 6w^2 - 4w + 1 - 6(w^3 - 3w^2 + 3w - 1)$
$+ 120w - 120 + 16 = 0$

$w^4 - 10w^3 + 24w^2 + 98w - 97 = 0$

Watch out Even though the x^2 term is 0, the equation is still a quartic. Find expressions for $\Sigma\alpha$, $\Sigma\alpha\beta$, $\Sigma\alpha\beta\gamma$ and $\alpha\beta\gamma\delta$.

If you expand one pair of roots you get $(2\alpha + 1)(2\beta + 1) = 4\alpha\beta + 2\alpha + 2\beta + 1$. Each original root will appear in three of the six possible pairs, giving you this expression.

If you expand one triple of roots you get $(2\alpha + 1)(2\beta + 1)(2\gamma + 1) = 8\alpha\beta\gamma + 4(\alpha\beta + \beta\gamma + \gamma\alpha) + 2(\alpha + \beta + \gamma + \delta) + 1$. There are four possible triples, with each root appearing three times and each pair of roots appearing twice.

Expand $(2\alpha + 1)(2\beta + 1)(2\gamma + 1)(2\delta + 1)$.

Problem-solving In many cases, it is quicker to use a substitution.

Multiply through by 16 to remove the fractions.

Expand and simplify.

Exercise 4E

1 The cubic equation $x^3 - 7x^2 + 6x + 5 = 0$ has roots α, β and γ. Find equations with roots:

a $(\alpha + 1)$, $(\beta + 1)$ and $(\gamma + 1)$ **b** 2α, 2β and 2γ

2 The cubic equation $3x^3 - 4x^2 - 5x + 1 = 0$ has roots α, β and γ. Find equations with roots:

a $(\alpha - 3)$, $(\beta - 3)$ and $(\gamma - 3)$ **b** $\frac{\alpha}{2}$, $\frac{\beta}{2}$ and $\frac{\gamma}{2}$

E/P **3** The cubic equation $x^3 - 3x^2 + 4x - 7 = 0$ has roots α, β and γ.
Without solving the equation, find the equation with roots $(2\alpha + 1)$, $(2\beta + 1)$ and $(2\gamma + 1)$.
Give your answer in the form $aw^3 + bw^2 + cw + d = 0$ where a, b, c and d are integers to be determined. **(5 marks)**

E/P **4** The cubic equation $x^3 + 4x^2 - 4x + 2 = 0$ has roots α, β and γ.
Without solving the equation, find the equation with roots $(2\alpha - 1)$, $(2\beta - 1)$ and $(2\gamma - 1)$.
Give your answer in the form $w^3 + pw^2 + qw + r = 0$ where p, q and r are integers to be found. **(5 marks)**

E/P **5** The cubic equation $3x^3 - x^2 + 2x - 5 = 0$ has roots α, β and γ.
Without solving the equation, find the equation with roots $(3\alpha + 1)$, $(3\beta + 1)$ and $(3\gamma + 1)$.
Give your answer in the form $aw^3 + bw^2 + cw + d = 0$ where a, b, c and d are integers to be determined. **(5 marks)**

P **6** The quartic equation $2x^4 + 4x^3 - 5x^2 + 2x - 1 = 0$ has roots α, β, γ and δ. Find equations with integer coefficients that have roots:

a 3α, 3β, 3γ and 3δ **b** $(\alpha - 1)$, $(\beta - 1)$, $(\gamma - 1)$ and $(\delta - 1)$

E/P **7** The quartic equation $x^4 + 2x^3 - 3x^2 + 4x + 5 = 0$ has roots α, β, γ and δ.
Without solving the equation, find equations with integer coefficients that have roots:

a 2α, 2β, 2γ and 2δ **(6 marks)**

b $(\alpha - 2)$, $(\beta - 2)$, $(\gamma - 2)$ and $(\delta - 2)$ **(6 marks)**

E/P **8** The quartic equation $3x^4 + 5x^3 - 4x^2 - 3x + 1 = 0$ has roots α, β, γ and δ.
Without solving the equation, find equations with integer coefficients that have roots:

a 3α, 3β, 3γ and 3δ **(6 marks)**

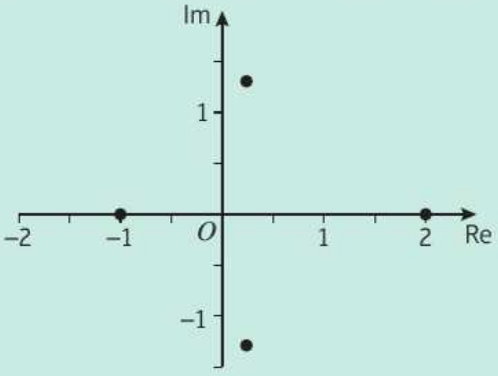
b $(\alpha + 1)$, $(\beta + 1)$, $(\gamma + 1)$ and $(\delta + 1)$ **(6 marks)**

Challenge

The quartic equation $2x^4 - 3x^3 + x^2 - 2x - 6 = 0$ has roots α, β, γ and δ.

a Find an equation with integer coefficients that has roots $(2\alpha + 1)$, $(2\beta + 1)$, $(2\gamma + 1)$ and $(2\delta + 1)$.

b The diagram shows the locations of the roots of the original equation on an Argand diagram:



Copy this diagram and sketch the approximate locations of the roots of your new equation.

Mixed exercise 4

1 The roots of a quartic equation $ax^4 + bx^3 + cx^2 + dx + e = 0$ are $\alpha = \frac{1}{5}$, $\beta = -\frac{2}{5}$, $\gamma = -\frac{3}{5}$ and $\delta = -\frac{1}{2}$
Find integer values for a, b, c, d and e.

E **2** The cubic equation $x^3 + px^2 + 37x - 52 = 0$ has roots α, β and γ.

a Write down the values of $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$, and express p in terms of α, β and γ. **(3 marks)**

b Given that $\alpha = 3 - 2i$, find the value of p. **(4 marks)**

E **3** The cubic equation $2x^3 + 5x^2 - 2x + q = 0$ has a root at $x = -2 + i$.

a Find the other two roots of the equation. **(4 marks)**

b Hence find the value of q. **(2 marks)**

E/P **4** The quartic equation $x^4 - 40x^3 + 510x^2 - 2200x + 1729 = 0$ has roots α, $\alpha + 2k$, $\alpha + 4k$ and $\alpha + 6k$ for some real constant k. Solve the equation. **(7 marks)**

E **5** Three of the roots of the quartic equation $24x^4 - 58x^3 + 17x^2 + dx + e = 0$ are $\frac{1}{2}$, $-\frac{1}{3}$ and 2.

a Find the fourth root. **(2 marks)**

b Find the values of d and e. **(4 marks)**

E/P **6** The equation $x^4 + 2x^3 + mx^2 + nx + 85 = 0$, $x \in \mathbb{C}$, $m, n \in \mathbb{R}$, has roots α, β, γ and δ.
Given that $\alpha = -2 + i$ and $\beta = \alpha^*$,

a show that $\gamma + \delta + 2 = 0$ and that $\gamma\delta - 17 = 0$. **(2 marks)**

b Hence find all the roots of the quartic equation and find the values of m and n. **(5 marks)**

c Show these roots on an Argand diagram. **(3 marks)**

E **7** A quartic equation $4x^4 - 16x^3 + 115x^2 + 4x - 29 = 0$ has roots α, β, γ and δ.

a Show that $2 - 5i$ is one root of the equation. **(3 marks)**

b Without solving the equation, find the other roots. **(5 marks)**

c Show these roots on an Argand diagram. **(3 marks)**

E **8** The roots of the equation $2x^3 - 5x^2 + 11x - 9 = 0$ are α, β and γ.

a Write down the values of $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$. **(1 mark)**

b Hence find the values of:

i $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ **(2 marks)**

ii $\alpha^2 + \beta^2 + \gamma^2$ **(2 marks)**

iii $(\alpha - 1)(\beta - 1)(\gamma - 1)$ **(3 marks)**

E **9** The roots of the equation $px^4 + 12x^3 + 6x^2 + 5x - 7 = 0$ are α, β, γ and δ.

a Given that $\alpha\beta\gamma\delta = -1$, write down the value of p. **(1 mark)**

b Write down the values of $\Sigma\alpha$, $\Sigma\alpha\beta$ and $\Sigma\alpha\beta\gamma$. **(1 mark)**

c Hence find the value of $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$. **(3 marks)**

E **10** The roots of the equation $5x^3 + cx + 21 = 0$ are α, β and γ.

a Given that $\alpha\beta + \beta\gamma + \gamma\alpha = -6$, write down the value of c. **(1 mark)**

b Write down values for $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$. **(1 mark)**

c Hence find the value of $(1 - \alpha)(1 - \beta)(1 - \gamma)$. **(3 marks)**

E/P **11** The cubic equation $2x^3 + 5x^2 + 7x - 2 = 0$ has roots α, β and γ.
Without solving the equation, find the equation with roots $(3\alpha + 1)$, $(3\beta + 1)$ and $(3\gamma + 1)$.
Give your answer in the form $pw^3 + qw^2 + rw + s = 0$ where p, q, r and s are integers to be found. **(5 marks)**

E/P **12** The quartic equation $6x^4 - 2x^3 - 5x^2 + 7x + 8 = 0$ has roots α, β, γ and δ.
Without solving the equation, find equations with integer coefficients that have roots:

a 2α, 2β, 2γ and 2δ **(6 marks)**

b $(3\alpha - 2)$, $(3\beta - 2)$, $(3\gamma - 2)$ and $(3\delta - 2)$ **(6 marks)**

Challenge

1 The cubic equation $x^3 + 4x^2 - 5x - 7 = 0$ has roots α, β and γ. Without solving the equation, find a cubic equation that has roots $\frac{1}{\alpha + 1}$, $\frac{1}{\beta + 1}$ and $\frac{1}{\gamma + 1}$

2 The cubic equation $x^3 + 2x^2 - 3x - 5 = 0$ has roots α, β and γ. Without solving the equation, find an equation that has roots $\alpha + \beta$, $\beta + \gamma$ and $\gamma + \alpha$.

3 The quartic equation $x^4 + 2x^2 - 5x + 2 = 0$ has roots α, β, γ and δ. By using a substitution, or otherwise, find an equation that has roots $\alpha^2 + 1$, $\beta^2 + 1$, $\gamma^2 + 1$ and $\delta^2 + 1$.

Summary of key points

1 If α and β are roots of the equation $ax^2 + bx + c = 0$, then:

- $\alpha + \beta = -\frac{b}{a}$
- $\alpha\beta = \frac{c}{a}$

2 If α, β and γ are roots of the equation $ax^3 + bx^2 + cx + d = 0$, then:

- $\alpha + \beta + \gamma = \Sigma\alpha = -\frac{b}{a}$
- $\alpha\beta + \beta\gamma + \gamma\alpha = \Sigma\alpha\beta = \frac{c}{a}$
- $\alpha\beta\gamma = -\frac{d}{a}$

3 If α, β, γ and δ are roots of the equation $ax^4 + bx^3 + cx^2 + dx + e = 0$, then:

- $\alpha + \beta + \gamma + \delta = \Sigma\alpha = -\frac{b}{a}$
- $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \Sigma\alpha\beta = \frac{c}{a}$
- $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = \Sigma\alpha\beta\gamma = -\frac{d}{a}$
- $\alpha\beta\gamma\delta = \frac{e}{a}$

Summary of key points

4 The rules for **reciprocals**:

- Quadratic: $\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}$
- Cubic: $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{\alpha\beta+\beta\gamma+\gamma\alpha}{\alpha\beta\gamma}$
- Quartic: $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta}=\frac{\alpha\beta\gamma+\beta\gamma\delta+\gamma\delta\alpha+\delta\alpha\beta}{\alpha\beta\gamma\delta}$

5 The rules for **products of powers**:

- Quadratic: $\alpha^n\times\beta^n=(\alpha\beta)^n$
- Cubic: $\alpha^n\times\beta^n\times\gamma^n=(\alpha\beta\gamma)^n$
- Quartic: $\alpha^n\times\beta^n\times\gamma^n\times\delta^n=(\alpha\beta\gamma\delta)^n$

6 The rules for **sums of squares**:

- Quadratic: $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$
- Cubic: $\alpha^2+\beta^2+\gamma^2=(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma\alpha)$
- Quartic: $\alpha^2+\beta^2+\gamma^2+\delta^2=(\alpha+\beta+\gamma+\delta)^2-2(\alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\beta\delta+\gamma\delta)$

7 The rules for **sums of cubes**:

- Quadratic: $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$
- Cubic: $\alpha^3+\beta^3+\gamma^3=(\alpha+\beta+\gamma)^3-3(\alpha+\beta+\gamma)(\alpha\beta+\beta\gamma+\gamma\alpha)+3\alpha\beta\gamma$